

Lecture 4

Protocols

Protocols

Network Protocol Overview

Network protocols define a common set of rules.

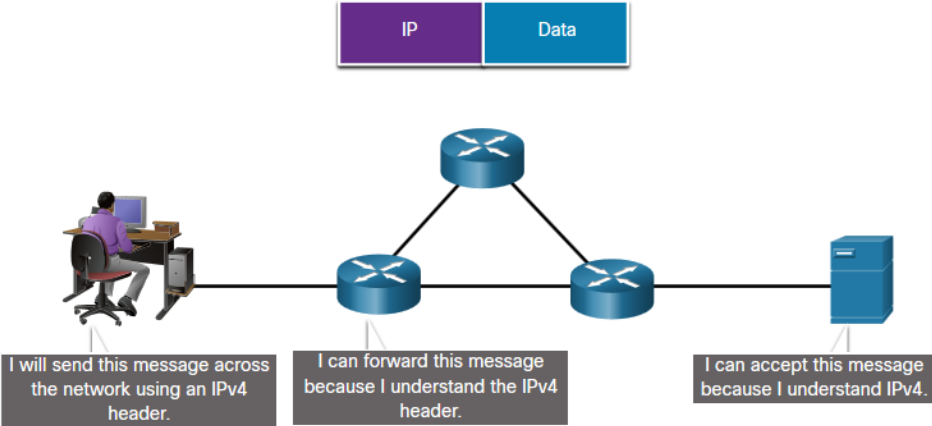
- Can be implemented on devices in:
 - Software
 - Hardware
 - Both
- Protocols have their own:
 - Function
 - Format
 - Rules

Protocol Type	Description
Network Communications	enable two or more devices to communicate over one or more networks
Network Security	secure data to provide authentication, data integrity, and data encryption
Routing	enable routers to exchange route information, compare path information, and select best path
Service Discovery	used for the automatic detection of devices or services

Protocols

Network Protocol Functions

- Devices use agreed-upon protocols to communicate .
- Protocols may have may have one or functions.

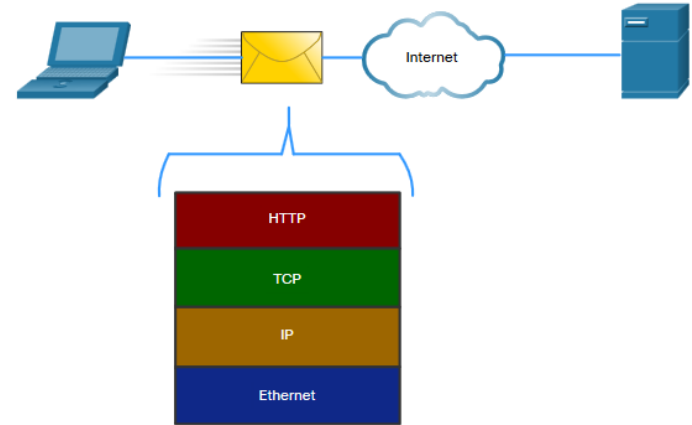


Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

Protocols

Protocol Interaction

- Networks require the use of several protocols.
- Each protocol has its own function and format.

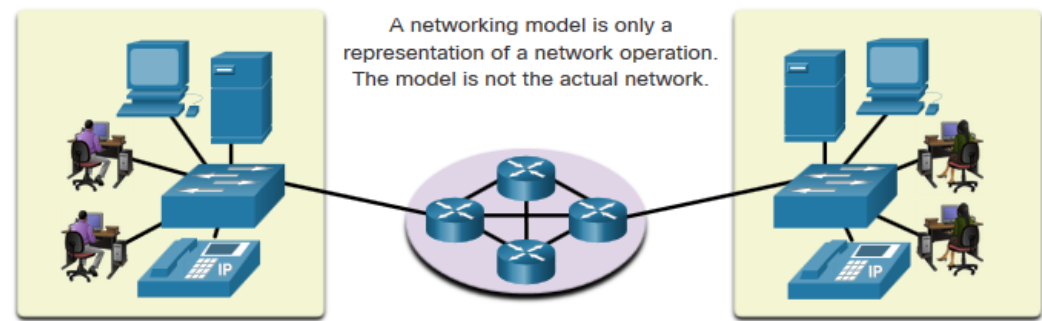


Protocol	Function
Hypertext Transfer Protocol (HTTP)	▪ Governs the way a web server and a web client interact ▪ Defines content and format
Transmission Control Protocol (TCP)	▪ Manages the individual conversations ▪ Provides guaranteed delivery ▪ Manages flow control
Internet Protocol (IP)	Delivers messages globally from the sender to the receiver
Ethernet	Delivers messages from one NIC to another NIC on the same Ethernet Local Area Network (LAN)

3.5 Reference Models

Reference Models

The Benefits of Using a Layered Model



Complex concepts such as how a network operates can be difficult to explain and understand. For this reason, a layered model is used.

Two layered models describe network operations:

- Open System Interconnection (OSI) Reference Model
- TCP/IP Reference Model

OSI Model	TCP/IP Protocol Suite	TCP/IP Model
Application	HTTP, DNS, DHCP, FTP	Application
Presentation		
Session		
Transport	TCP, UDP	Transport
Network	IPv4, IPv6, ICMPv4, ICMPv6	Internet
Data Link	Ethernet, WLAN, SONET, SDH	Network Access
Physical		

The Benefits of Using a Layered Model (Cont.)

These are the benefits of using a layered model:

- Assist in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
- Foster competition because products from different vendors can work together
- Prevent technology or capability changes in one layer from affecting other layers above and below
- Provide a common language to describe networking functions and capabilities

The OSI Reference Model

OSI Model Layer	Description
7 - Application	Contains protocols used for process-to-process communications.
6 - Presentation	Provides for common representation of the data transferred between application layer services.
5 - Session	Provides services to the presentation layer and to manage data exchange.
4 - Transport	Defines services to segment, transfer, and reassemble the data for individual communications.
3 - Network	Provides services to exchange the individual pieces of data over the network.
2 - Data Link	Describes methods for exchanging data frames over a common media.
1 - Physical	Describes the means to activate, maintain, and de-activate physical connections.

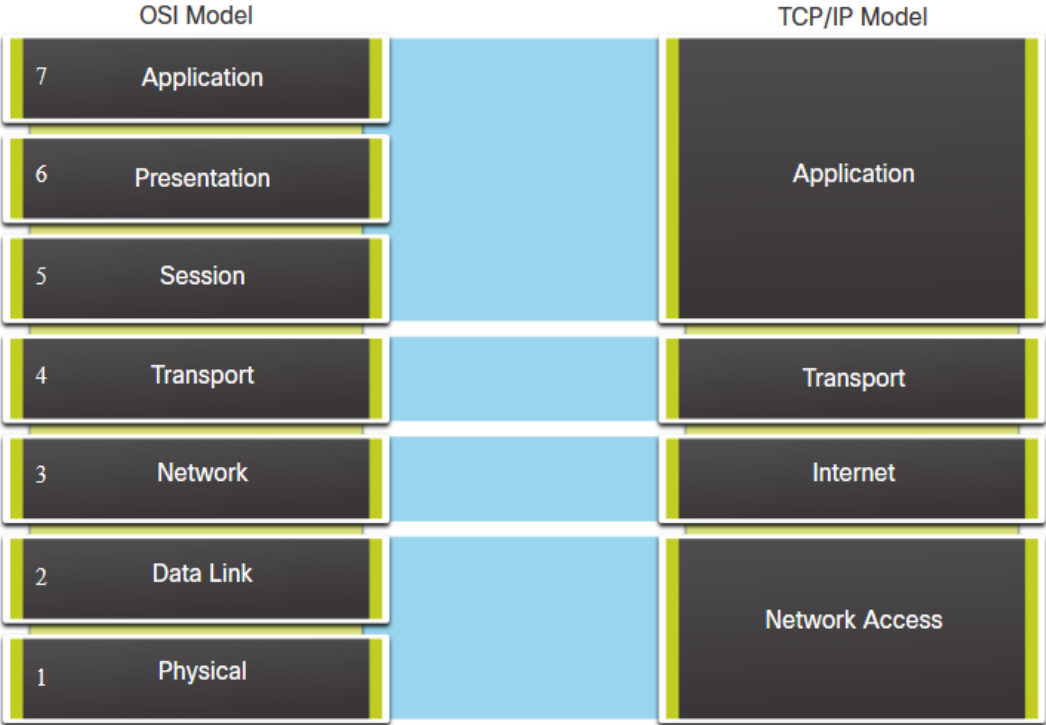
Reference Models

The TCP/IP Reference Model

TCP/IP Model Layer	Description
Application	Represents data to the user, plus encoding and dialog control.
Transport	Supports communication between various devices across diverse networks.
Internet	Determines the best path through the network.
Network Access	Controls the hardware devices and media that make up the network.

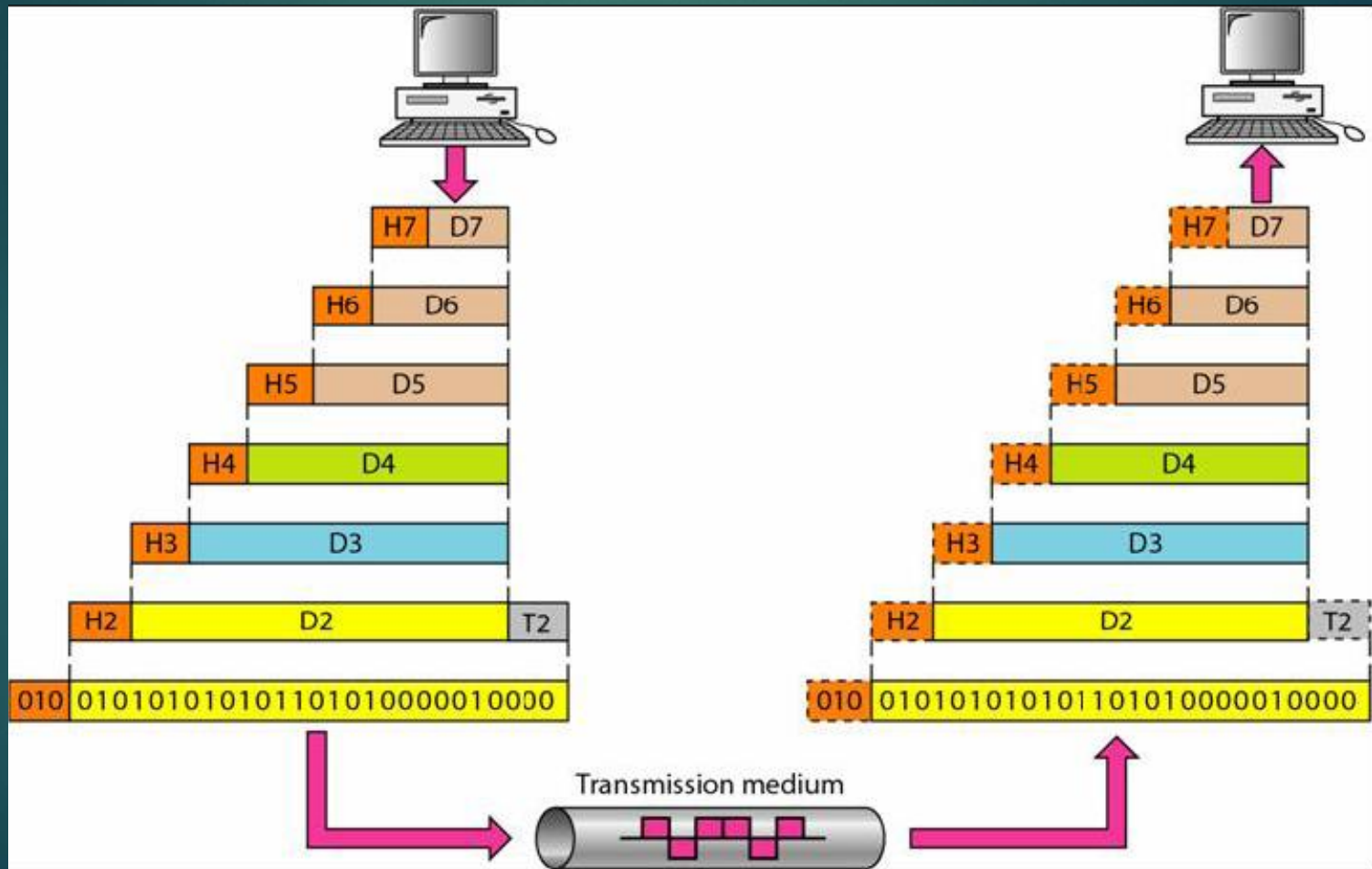
Reference Models

OSI and TCP/IP Model Comparison

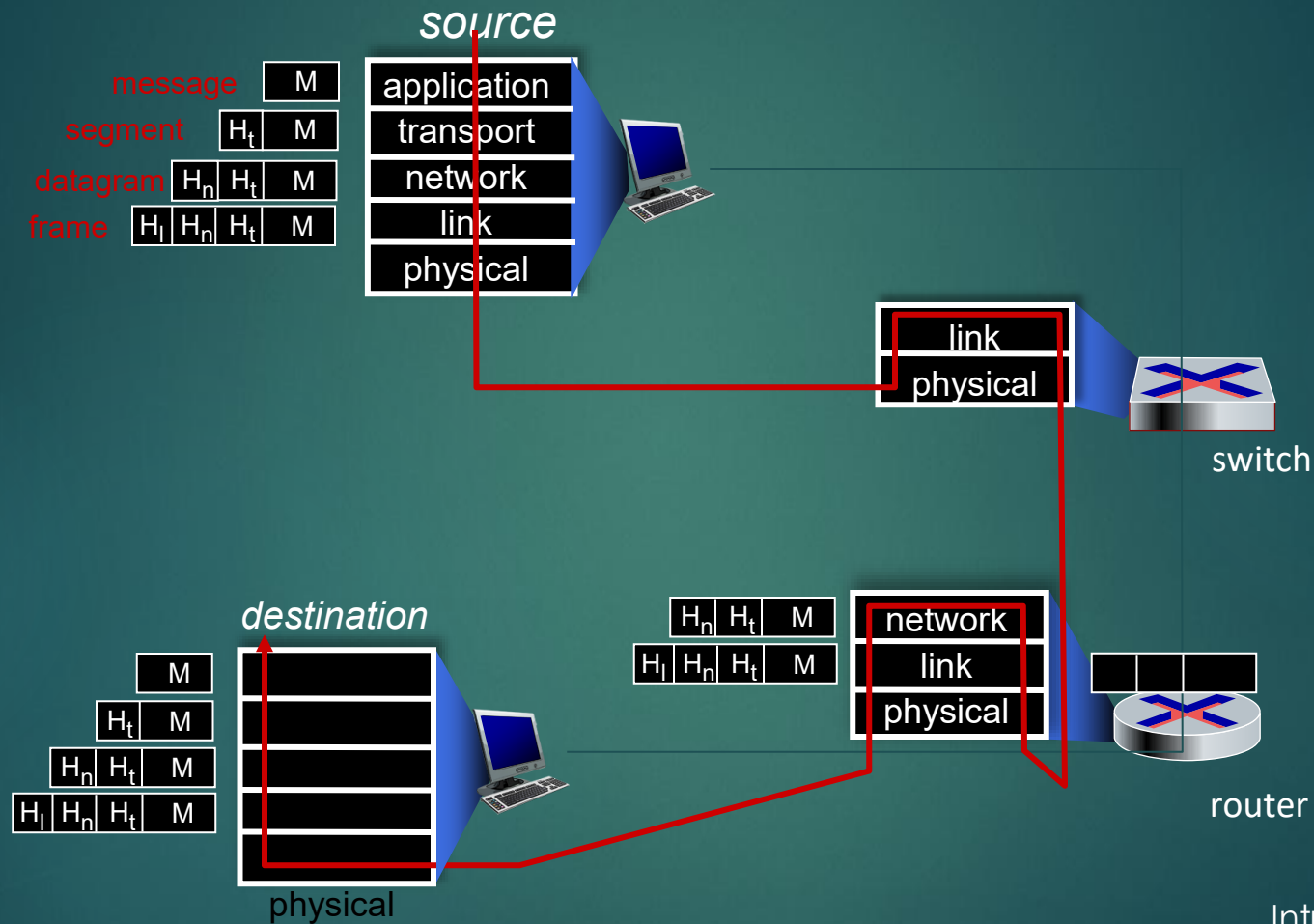


- The OSI model divides the network access layer and the application layer of the TCP/IP model into multiple layers.
- The TCP/IP protocol suite does not specify which protocols to use when transmitting over a physical medium.
- OSI Layers 1 and 2 discuss the necessary procedures to access the media and the physical means to send data over a network.

An exchange using the OSI model

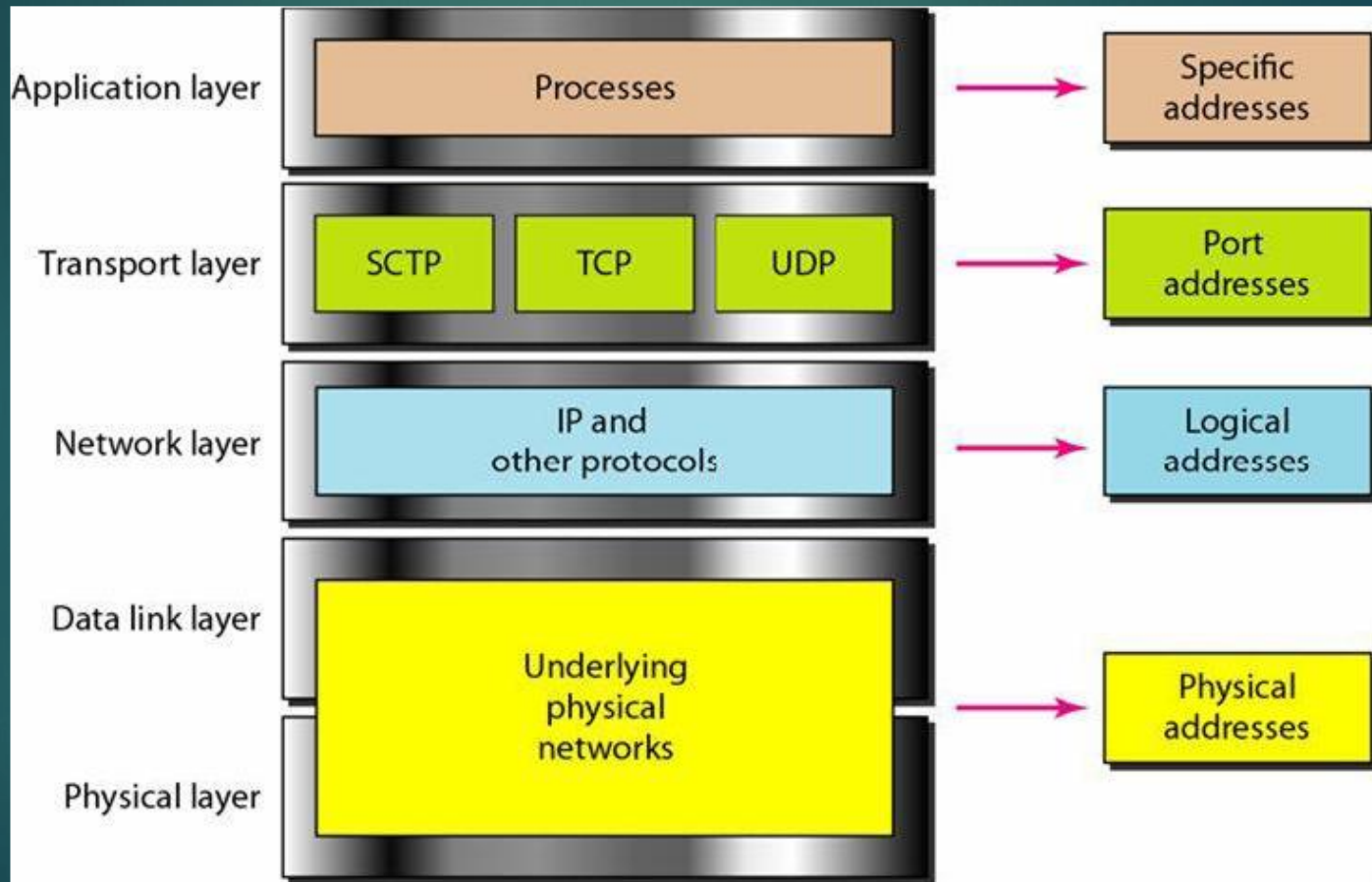


Encapsulation: an end-end view



Addressing

- ▶ Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.



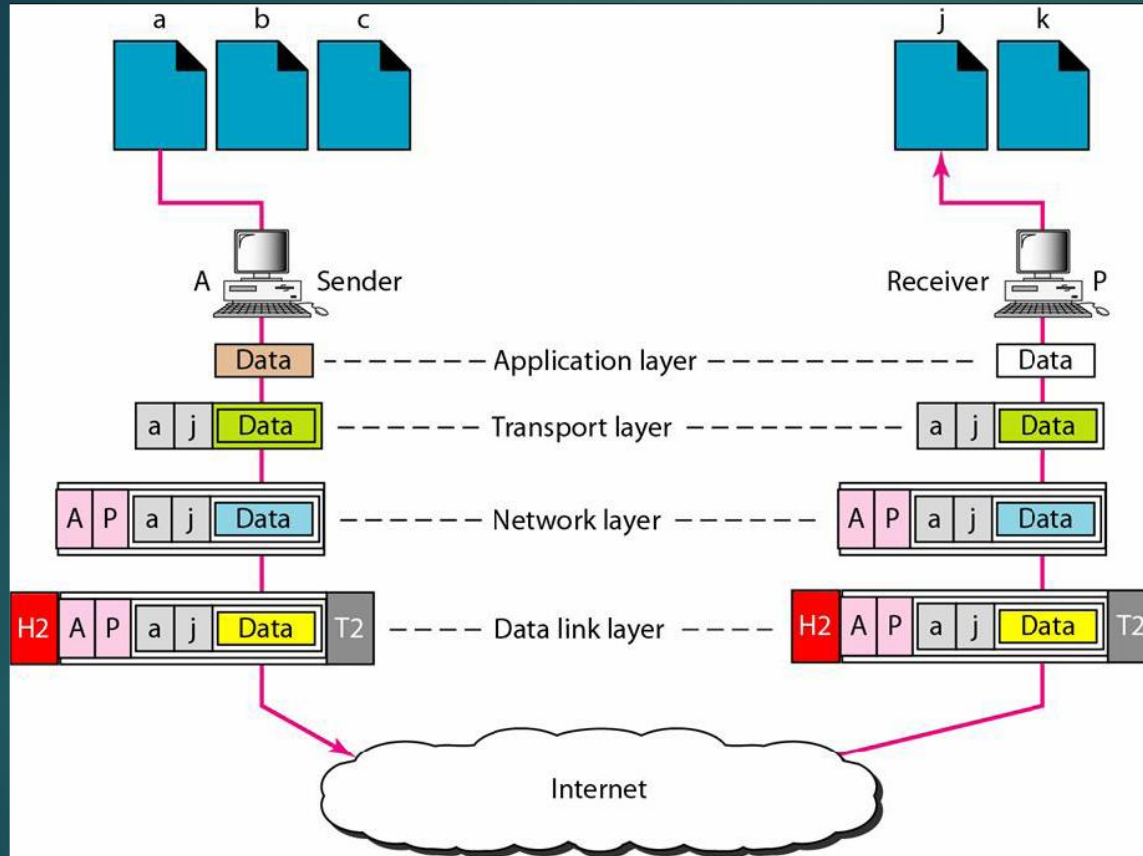
Physical Address

- ▶ Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits
 - ▶ Example 07:01:02:01:2C:4B

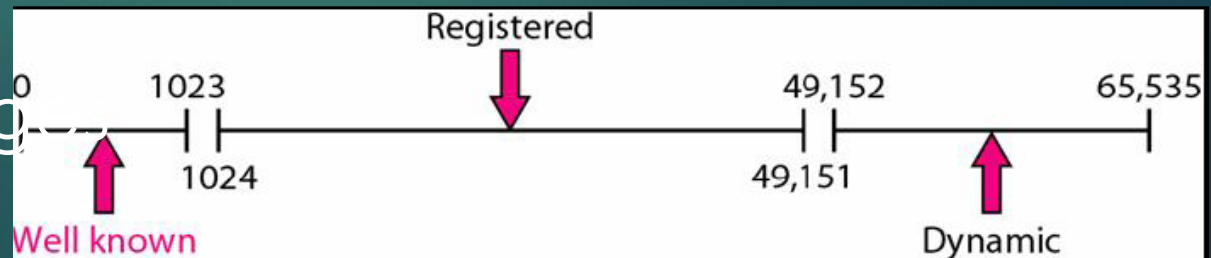
IP Address (Logical Address)

- ▶ An IP address is a numeric identifier assigned to each machine on an IP network. It designates the specific location of a device on the network.
- ▶ IP addressing was designed to allow hosts on one network to communicate with a host on a different network regardless of the type of LANs the hosts are participating in.

Port addresses



Port Number Range



Specific addresses

- ▶ User friendly address such as www.kku.ac.th (URL) or your email address

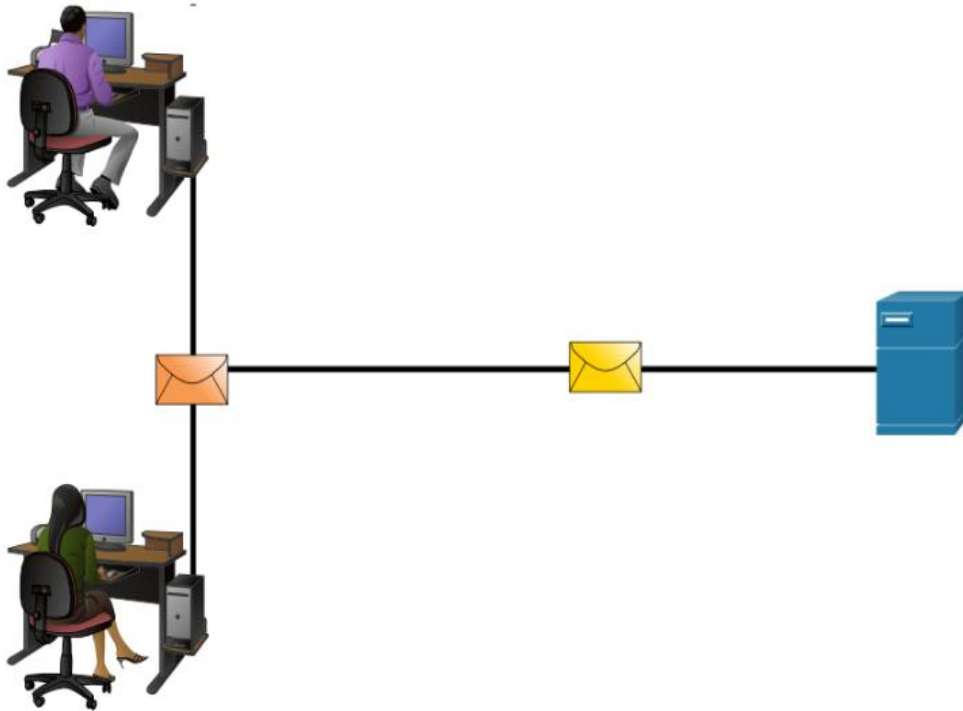
Domain Name System

- ▶ Service and system used to translate IP addresses to human readable "domain names"
- ▶ Example.org has address 202.95.12.34
- ▶ 93.184.216.34 has hostname example.org
- ▶ Records have different types for different purposes
- ▶ Commonly runs over TCP & UDP port 53

3.6 Data Encapsulation

Data Encapsulation

Segmenting Messages



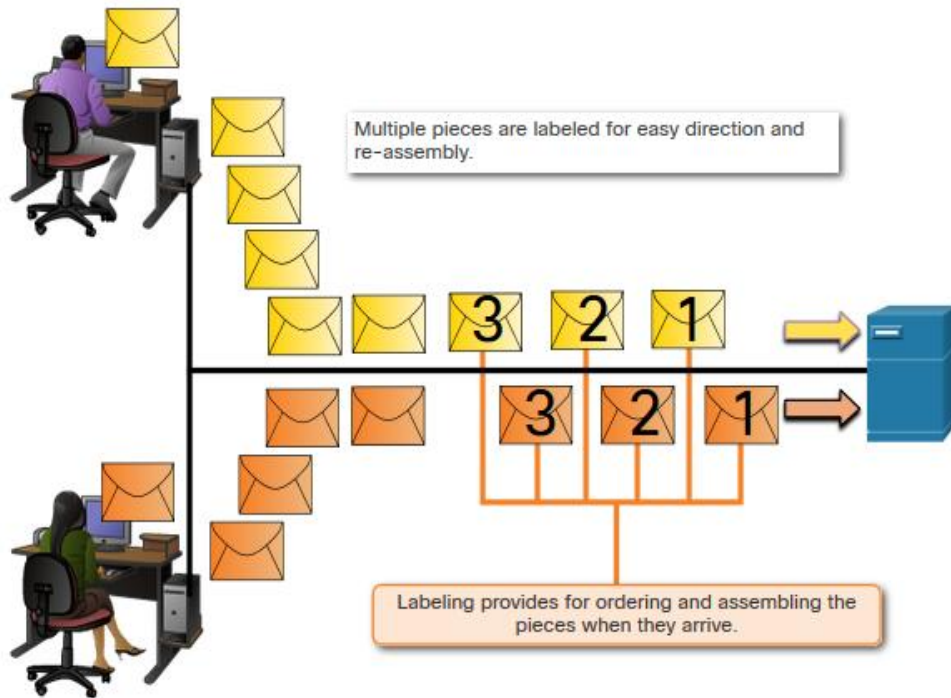
Segmenting is the process of breaking up messages into smaller units. Multiplexing is the processes of taking multiple streams of segmented data and interleaving them together.

Segmenting messages has two primary benefits:

- **Increases speed** - Large amounts of data can be sent over the network without tying up a communications link.
- **Increases efficiency** - Only segments which fail to reach the destination need to be retransmitted, not the entire data stream.

Data Encapsulation

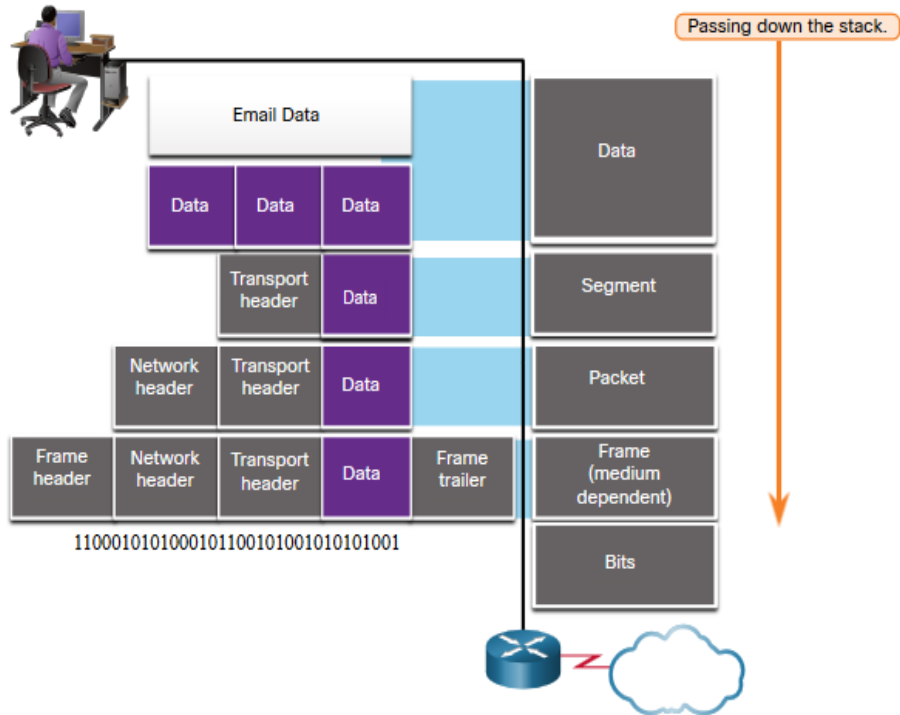
Sequencing



Sequencing messages is the process of numbering the segments so that the message may be reassembled at the destination.

TCP is responsible for sequencing the individual segments.

Data Encapsulation Protocol Data Units



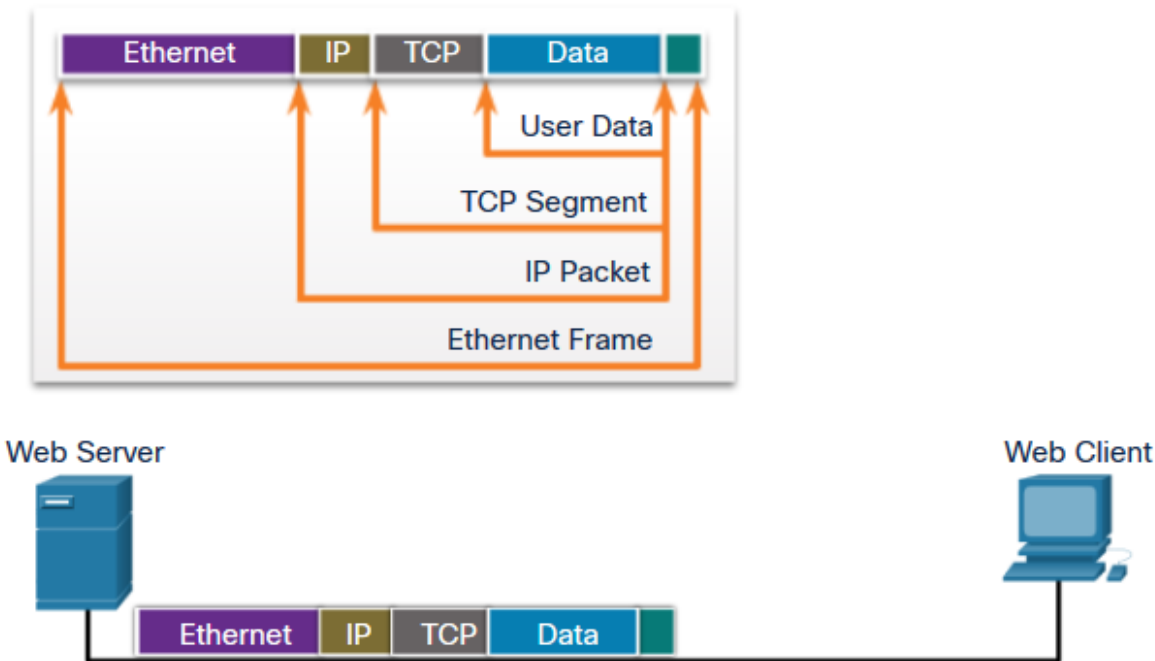
Encapsulation is the process where protocols add their information to the data.

- At each stage of the process, a PDU has a different name to reflect its new functions.
- There is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite.
- PDUs passing down the stack are as follows:
 1. Data (Data Stream)
 2. Segment
 3. Packet
 4. Frame
 5. Bits (Bit Stream)

Data Encapsulation

Encapsulation Example

- Encapsulation is a top down process.
- The level above does its process and then passes it down to the next level of the model. This process is repeated by each layer until it is sent out as a bit stream.



Data Encapsulation

De-encapsulation Example

- Data is de-encapsulated as it moves up the stack.
 - When a layer completes its process, that layer strips off its header and passes it up to the next level to be processed. This is repeated at each layer until it is a data stream that the application can process.
1. Received as Bits (Bit Stream)
 2. Frame
 3. Packet
 4. Segment
 5. Data (Data Stream)

